

Multi-tone Configurable Alarm

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Presented here is a multi-tone configurable alarm that can be activated by temperature and other physical parameters. Its output tone can be selected through different switch combinations. Such an alarm is highly desirable when multiple alarms are present in the same area. Different tones can be selected for different alarms so that you can easily know which alarm has triggered. The alarm in this circuit is activated by temperature.

Circuit and working

CD4060 (IC1) is a CMOS integrated circuit that contains an oscillator and a 14-stage ripple counter/divider. It has three oscillator terminals (Φ_1 , Φ_0 and Φ_2), ten buffered outputs of the counter (Q4-Q10 and Q12-Q14) and an active 'high' master reset. It is used in this project to produce ten different

square waves at the same time. These square waves can be mixed to produce different output tones using switches (S1-S10).

The oscillator frequency mainly depends on the value of components C3, R2, R1 and VR1. If needed, oscillator frequency can be trimmed using potmeter VR1. Trimming is useful to obtain different clock frequencies when several alarms are in use near each other. If trimming is not needed, VR1 can be replaced with a fixed-value resistor.

With increasing temperature, resistance of thermistor NTC1 decreases and the biasing voltage for transistor



PARTS LIST

Semiconductors:

- IC1 - CD4060/74HCT4060 counter-cum-oscillator
- LED1, LED2 - 5mm LED
- T1, T3 - BC337 npn transistor
- T2 - BD139 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 220-kilo-ohm
- R2 - 10-kilo-ohm
- R3 - 5.6-kilo-ohm
- R4, R16, R17 - 1-kilo-ohm
- R5-R14 - 10-kilo-ohm
- R15 - 10-ohm
- VR1 - 100-kilo-ohm potmeter
- VR2 - 22-kilo-ohm potmeter

Capacitors:

- C1 - 100pF ceramic disk
- C2 - 10nF ceramic disk
- C3 - 120pF ceramic disk
- C4, C5 - 0.33 μ F ceramic disk
- C6 - 100 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal block
- LS1 - 8-ohm, 0.5W speaker
- NTC1 - 100-kilo-ohm NTC thermistor
- S1-S10 - On/off switch

Test Points

Test point	Details
TP0	GND
TP1	5V
TP2	Low when NTC1 is hot
TP3	Square wave (frequency depends on VR1)

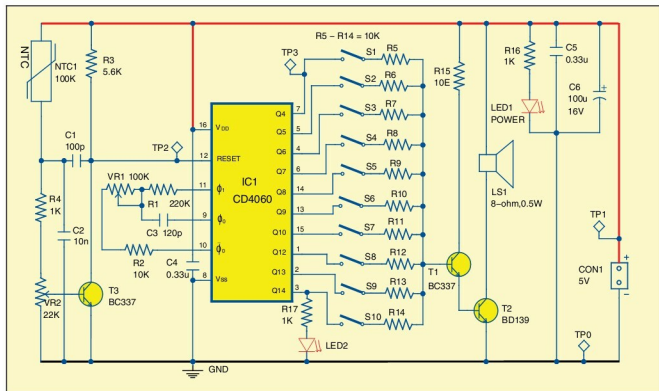


Fig. 1: Circuit diagram of multi-tone configurable alarm